

ReSolve: Verified Mechanical CAD Edits and Linked Vector Drawings

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Abstract

001 Editing a mechanical drawing is useful only when the
002 underlying part changes as requested and the delivered
003 views remain consistent. We study a CAD-first work-
004 flow that predicts a validated program patch, executes
005 the edited CadQuery model, and exports linked SVG
006 views. Our main experiment uses published Bench-
007 CAD edit pairs with a local family- and template-
008 component-disjoint train/validation/test split. A fresh
009 1.5B QLoRA model is trained on 492 retained edits
010 and evaluated on all 126 retained held-out cases, us-
011 ing source-program equality and the author’s released
012 STEP solids. The adapter yields 77/126 applicable
013 patches and 12 exact target programs; 12 of 125 valid
014 released STEP references are matched strictly. A sub-
015 sequent three-epoch H100 LoRA run on the same re-
016 tained split produces 61/126 applicable patches and
017 15/125 strict solid matches. We report executable
018 solids and volume agreement separately, with a format-
019 relaxed base-model control. A smaller, explicitly local-
020 ized pilot provides 150 verified edit pairs and paired
021 drawing sheets. Separate deterministic truss and plate
022 controls demonstrate axial FEM equilibrium and clear-
023 ance violations; those results are not attributed to
024 the learned CAD model. The study measures code-
025 conditioned editing and linked view generation, not
026 drawing perception or general engineering safety. The
027 release lacks physical operating conditions; a separate,
028 explicitly scenario-defined 3D elasticity audit checks six
029 illustrated source/target/prediction triples after train-
030 ing.

031 1. Introduction

032 A request to remove a bracket flange, enlarge a bore or
033 add a shaft slot must change more than a caption. The
034 edited part should preserve unrelated features, remain
035 executable, and produce coherent views. If the request

concerns structural performance, it additionally needs
material, units, supports, loads and a declared phys-
ical model. Shape validity alone cannot supply those
assumptions.

This paper focuses on mechanical CAD edits and
their linked vector drawings. Figure 1 illustrates two
published reference pairs: removing a flange while re-
taining the twisted web and first-flange holes, and
adding an obround blind slot to a stepped shaft. These
require compound three-dimensional geometry rather
than a symbol substitution. The figure shows released
references, not model predictions.

We implement a CAD-first edit-execute-check
route. A language model reads an executable pro-
gram and the user’s instruction and returns a source-
line patch. A validating executor applies the patch to
the original program. CadQuery builds the resulting
solid; orthographic and isometric SVG views are ex-
ported from that solid. Independent scoring compares
the executed result with the author’s STEP reference.
A malformed patch, invalid solid and valid but incor-
rect edit are distinct outcomes.

Our main data comes from an online release rather
than only locally manufactured requests: BenchCAD’s
published edit pairs [14]. Because its edit subset is
originally a held-out benchmark, using it for training
changes the protocol. We explicitly construct a local
split before training, hold out complete family/tem-
plate components and start from fresh base weights.
We do not report an official BenchCAD leaderboard
result.

Contributions and scope.

- A reproducible local training/evaluation protocol
over published mechanical edit pairs, with pinned
provenance, family/template isolation, disclosed to-
ken exclusions and every retained test case scored.
- A validated CAD-program patch interface and de-
terministic linked SVG export, evaluated with both
program and executed-solid agreement. The learned

model receives code; it is not a drawing-perception model.

- A separate 150-pair hard-part pilot with before/after STEP solids and four-view drawing sheets, plus a transport control showing why patch-format gains must not be equated with engineering reasoning.
- Explicit separation of mechanical geometry checks, deterministic structural FEM equilibrium and plate clearance rules. No mechanical safety or broad method-novelty claim follows from these small, single-seed experiments.

2. Related work

Executable CAD and editing. DeepCAD [12] and Text2CAD [5] model parametric construction histories. CAD-Editor [13] studies instruction-based sequence editing. BenchCAD [14] supplies executable mechanical programs and published edit pairs, while neuralCAD-Edit [8] evaluates expert-requested edits. Our experiment reuses released supervision; a program-patch interface and execution check are implementation choices, not evidence that CAD editing itself is new.

Drawings and vector graphics. Drawing2CAD [9] links vector views to CAD histories. DeepSVG [1] and StarVector [10] model vector graphics, and SVGE-ditBench [7] benchmarks instruction-based SVG edits. Our learned experiment conditions on CAD code and exports SVG after execution. It therefore does not establish SVG-to-CAD extraction, image understanding, or robust annotation recovery from manufacturing sheets.

Physical evaluation. FEM-Bench [6] tests scientific code generation. CADEngBench [11] evaluates functional CAD edits and matched linear-static FEA. Geometry and FEA are complementary: the latter requires an analysis contract. We retain an explicitly specified axial-truss control but do not fabricate loads or materials for BenchCAD parts. A public runnable CADEngBench release was not verified in this work, so its reported FEA results are related work rather than our reproduced benchmark.

3. Mechanical edit–execute–check workflow

3.1. Program and patch contract

Let C be a CadQuery program and u an edit instruction. The model predicts a patch p , and the executor produces $\hat{C} = \mathcal{E}(C, p)$. Each patch operation contains

a zero-based original line index, a deletion count and a list of replacement lines. All coordinates refer to the original source, not an intermediate revision. Operations must be ordered, non-overlapping, within bounds and well typed. Parsing the executed program provides a second syntax check.

Training targets are line diffs between the published source and edited programs after removing comments and empty lines. Comment normalization avoids supervising changes to the upstream parameter-comment block as geometric edits. A round-trip check requires that applying every target patch reproduces its normalized target exactly. The online instructions are preserved; source-line answers are not added to the prompt.

3.2. Execution and vector views

CadQuery executes the patched program to a boundary-representation solid $\hat{B}$. The scorer requires valid geometry, at least one solid and positive volume. Downloaded programs run through a restricted executor with permitted imports, no private-attribute access or file-I/O builtins, and a 45-second per-case subprocess budget. This restriction is defense in depth, not a general Python sandbox. Unsupported constructs, invalid programs and timeouts are recorded separately.

For the verified drawing pilot, front, top, right and isometric SVG projections of both original and edited solids are saved alongside STEP exports. A before/after sheet binds these views to the request and geometry measurements. The projections share a three-dimensional source; they are not independently generated views. They are CAD-derived line drawings with edit annotations, not fully dimensioned production blueprints. Automatic projection scaling can differ between views, so rendered pixel size is not a physical measurement.

3.3. Independent solid scoring

The online test reference is the author’s released STEP solid B^* , pinned by content hash. We report volumetric intersection-over-union,

$$\text{IoU}(\hat{B}, B^*) = \frac{V(\hat{B} \cap B^*)}{V(\hat{B}) + V(B^*) - V(\hat{B} \cap B^*)}. \quad (1)$$

Our local thresholds are 0.95 for broad volume agreement and 0.99999 for strict agreement. Neither is presented as an official BenchCAD acceptance rule. Exact normalized-program AST equality is complementary: IoU may overlook a small but important feature, and multiple programs can produce the same shape. The current scorer does not certify feature-specific tolerances, preserved constraints, assembly joints, dimen-

Published BenchCAD source / reference edit pairs

twisted bracket | Remove the second flange; retain the twisted web and first-flange holes.

SOURCE

RELEASED TARGET



stepped shaft | Add a centered blind obround slot to the upper shaft face.

SOURCE

RELEASED TARGET



Reference illustrations only. No stress or FEM result is inferred from these shapes. BenchCAD, CC-BY-4.0.

Figure 1. Published mechanical CAD edit pairs held out in the local protocol. Top: remove the perpendicular end flange while preserving the twisted web and the first-flange holes. Bottom: add a centered blind obround slot to the upper cylindrical shaft section. These are source/released-target illustrations, not learned model predictions. Both views are exported from solids; shape appearance supplies no stress or FEM result; the separate FEM audit concerns the six actual predictions in the supplement.

sions, standards compliance or manufacturing suitability.

Verification is post-edit evaluation. The QLoRA loss is supervised token prediction, not FEM or solver feedback; the experiment does not evaluate iterative repair after a failed check.

4. Online dataset and split protocol

Published pairs. We use the `edit-bench` configuration of `BenchCAD/BenchCAD`, pinned at revision `5919f578ab09ec283603a082fab07c7639ab56eb`. It contains 748 instruction/source/target pairs with original and target STEP data [14]. The dataset declares CC-BY-4.0. Our downloaded parquet checksum and source-record identifiers are retained. Pipe, duct, fitting, manifold, valve, hose, resistor, circuit and PCB family names are excluded, removing 38 records and leaving 710 mechanical edit pairs.

Isolation before training. A complete mechanical family is one initial group. Families whose source or target ASTs share a template after masking numeric literals are joined into a component. Hash-ranked components

are assigned approximately 70/10/20 to training, validation and test before normalization, patches or augmentation. This gives 533/50/127 pairs. Family and component identities are disjoint across splits; no target from a held-out family enters fine-tuning. Template isolation is conservative but does not prove all cross-dataset geometric ancestry is disjoint. The fresh run does not load the earlier mechanical adapter.

Context-limited coverage. We retain a pair only when its full chat, including target, fits 3,072 tokens and its target patch fits 512 tokens. The rule is applied once before baseline evaluation and training. It retains 492/44/126 train/validation/test pairs, excluding 48 records across all splits. No example is silently truncated. The one excluded test pair and all excluded training examples remain identified in the run manifest. Results apply to the retained slice rather than the entire release.

Protocol change. The author labels this edit subset a held-out benchmark. We deliberately re-partition it into local training and evaluation data to satisfy the present study’s training objective. This invali-

214 dates claims of official benchmark-test generalization
215 or direct leaderboard comparability. The local held-
216 out components remain unused for optimization and
217 checkpoint selection.

218 Reference-quality limits. Readback finds one invalid
219 rivet STEP among the 126 retained test references.
220 We also identify a concrete instruction/target con-
221 flict: `topup_stepped_shaft_top_slot_rev` requests
222 cutting a blind slot, but its source already has the slot
223 and its target removes the operation. Published la-
224 bels are retained without silently repairing them; this
225 case and its source diff are recorded. The experiment
226 measures agreement with released targets, not certi-
227 fied compliance with every natural-language request.
228 We do not infer that every reverse-edit record has this
229 defect.

230 Earlier drawing pilot. A separate sample of 2,000
231 BenchCAD code-generation parts yielded 150 verified
232 local numeric-edit pairs from 76 hard parts and 40 fam-
233 ilies, partitioned 108/22/20 by family. It requires at
234 least eight CAD calls and compound geometry, and
235 excludes external edit-program templates after mask-
236 ing literals. Chamfer, hole, extrusion and fillet edits
237 contribute 62, 48, 24 and 16 pairs. The sample exports
238 904 referenced SVG views and 150 before/after sheets.
239 Its instructions disclose the edit line and numeric an-
240 swer; it is a transport feasibility pilot, not the main
241 online benchmark.

242 5. Online editing experiments

243 Training. We start from a fresh 4-bit Qwen2.5-
244 Coder-1.5B-Instruct conversion [4] and train a QLoRA
245 adapter [3] using MLX on an Apple M5 with
246 16 GB memory. The model conversion revision is
247 `b3252a2f97102b1fb1571fec2c9b27219a8536be`. Seed
248 17, batch size one, learning rate 10^{-4} and the last eight
249 transformer blocks are fixed. Adapters have rank 8,
250 scale 20, no dropout, and cover attention and MLP
251 linear projections in those blocks. Prompt tokens are
252 masked from the loss. We run 492 optimizer steps, cor-
253 responding to one pass of the retained training rows,
254 and use the final checkpoint. Validation is logged, not
255 used to select among test-informed alternatives. Saved
256 adapters are reloaded before evaluation.

257 Evaluation and metrics. Base and trained models re-
258 ceive the same original programs, published instruc-
259 tions and patch contract. Greedy decoding is capped
260 at 512 new tokens. All 126 retained test records are

261 evaluated in the same fixed order. Applicability re-
262 quires a valid patch and syntactically valid program;
263 AST equality requires the executed program to match
264 the published target’s normalized syntax. Solid scoring
265 executes the actual prediction and compares it with the
266 released STEP reference. No gold edits repair a model
267 prediction.

268 Transport control. The base model frequently emits
269 a bare list of edits rather than the required object con-
270 taining `edits`. We therefore add a deterministic diag-
271 nostic that wraps a syntactically valid bare list while
272 preserving every operation. It makes 56 of 126 base
273 predictions applicable, but none matches the target
274 AST. This control distinguishes one schema mismatch
275 from the remaining content and coordinate errors; it
276 does not rewrite geometry, infer missing operations or
277 correct indexing.

278 Results and operational coverage. The trained model
279 produces 77/126 applicable patches and 12/126 exact
280 target ASTs. Of 125 independently valid STEP refer-
281 ences, 39 predictions build valid solids, 18 reach $\text{IoU} \geq 0.95$,
282 and 12 reach strict $\text{IoU} \geq 0.99999$ (9.6%).
283 The bare-list base control builds 3 solids, with 1 broad
284 and 0 strict matches. Thus the experiment exposes a
285 substantial remaining gap: most held-out edits fail to
286 produce the reference geometry, particularly beyond
287 literal-parameter changes. Independent reference read-
288 back finds 125/126 valid target solids. One released
289 rivet reference is invalid and is excluded from the geo-
290 metric success denominator, while all task outputs re-
291 main reported. For the trained run, 39 cases are geo-
292 metrically scored, 37 have execution errors and 49 have
293 invalid patches; there are 0 timeouts and 0 Boolean-
294 comparison errors. The observed code failures include
295 invalid feature operations and damaged source coord-
296 inates; they are distinct from valid solids with low
297 IoU . We keep base and trained outputs, patch errors,
298 executed-solid measurements, reference hashes and cat-
299 egory breakdowns. One seed and one model size cannot
300 establish a robust scaling law or broad superiority over
301 CAD agents. No official benchmark claim or statistical
302 population failure rate is inferred from this local slice.

303 Additional H100 run. After the local experiment,
304 available server compute supports a separately re-
305 ported CUDA run. A fresh unquantized BF16
306 Qwen2.5-Coder-1.5B-Instruct model is trained on the
307 identical 492 retained training rows for three epochs
308 (1,476 updates) on one NVIDIA H100 NVL. PEFT
309 adapters use rank 8, alpha 160 (scale 20), no dropout
310 and the same seven attention/MLP projections in the

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
Base, strict interface	0	0	0	0
Base, bare-list transport control	56	0	1	0
Fresh online-pair QLoRA	77	12	18	12

Table 1. Local online-pair benchmark: counts out of 126. All systems use the same family/template holdout. The transport control only wraps a bare JSON edit list into the required object, without changing coordinates or content. Timeouts and comparison errors remain separately disclosed; they are not measured incorrect geometry. This is a local split, not an official BenchCAD score.

Category	Cases	Target AST	Strict match
T1	31	11	11
T2	18	1	1
T3	18	0	0
T4	34	0	0
T5	25	0	0

Table 2. Trained results by published edit category. Category IDs are retained from the source; counts are not pooled across different evaluation contracts.

last eight blocks. Seed, batch size, learning rate, prompt masking, context limit and greedy generation limit remain unchanged. We reload the final adapter and evaluate all 126 cases; no test-based checkpoint selection is performed. Different precision, software implementations and epoch counts prevent interpreting the comparison as an isolated hardware effect or an independent preregistered confirmation.

The final CUDA adapter yields 61/126 applicable patches and 16/126 target AST matches. Against 125 valid released STEP solids, 36 predictions execute, 19 meet broad IoU and 15 meet strict IoU (12.0%). There are 65 invalid patches, 24 execution errors, 0 timeouts and 0 comparison errors; the same invalid released reference is excluded from the geometry denominator.

Localized drawing pilot. The earlier 100-step adapter retained 92 training, 22 validation and 12 test examples after context filtering. All 12 held-out exact patches execute and match their references. Under a unique-substring line parser the base model already matches 11/12 solids. That result mainly improves transport formatting and cannot support a claim of advanced CAD reasoning. The online experiment uses published instructions without line hints, a fresh adapter and its own held-out families.

6. Separate structural and clearance controls

The full CAD benchmark above checks mechanical shape. A separate six-case FEM audit is described in the supplement. We retain a separately identified deterministic SVG route to show what physical or man-

ufacturing checks require when a drawing does supply a contract. These synthetic controls are not training data or accuracy evidence for the learned BenchCAD model.

Axial trusses. An importer reads supported untagged truss geometry, section dimensions, material notes, nodal loads and supports. A bounded request interpreter generates validated SVG tree patches. Re-importing the actual patched SVG determines member lengths, orientations and axial areas. Linear-elastic pin-jointed elements use EA/L stiffness; assembled displacements, stresses and reactions are recomputed. The numerical verdict checks equilibrium under small-displacement and nodal-load assumptions, not buckling, fatigue, joints or material strength.

Plate clearances. A separate importer recovers a rectangular boundary and circular-hole centres, diameters, thickness and stated limits. It computes overlap, edge distance and ligament. A faithful hole-enlargement request can reduce clearance below the printed minimum. A geometric violation is an engineering consequence of a correctly executed request, not necessarily an edit error; no continuum plate stress follows from a clearance score.

For example, increasing all truss member depths from 20 to 40 mm changes the recovered area and reduces peak axial stress from 11.861 to 5.930 MPa under the same load. Enlarging plate holes from 10 to 16 mm at fixed centres 20 mm from a boundary reduces edge clearance from 15 to 12 mm and fails a 15 mm rule. These are distinct checks with explicitly stated assumptions. The saved audit and replayed source/edited SVGs remain separate from the new online training run.

When FEM is applicable. A stress, stiffness, buckling, thermal or contact request needs physical units, constitutive properties, loads and boundary conditions, followed by solver and mesh checks. The inspected BenchCAD edit pairs do not provide this analysis contract.

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
CUDA base	0	0	0	0
CUDA base, list wrapper	47	0	0	0
CUDA final LoRA	61	16	19	15

Table 3. Additional three-epoch H100 experiment. Counts cover the same 126 retained tasks and 125 valid references, using the unchanged local geometry thresholds. Both base and final model are decoded with the CUDA implementation; the wrapper control preserves predicted edit contents.

Route	Edits	Outcome
Axial trusses	20,008	Equilibrium passes
Perforated plates	19,992	699 clearance violations

Table 4. Historical structural pipeline audit. All 40,000 supported edits match their stored target trees. These counts describe a deterministic synthetic pipeline, not a learned CAD benchmark. The plate violations are faithful edits flagged by the geometric checker.

We retain not evaluated for the full released benchmark and distinguish the added six-case, synthetic-load FEM audit from validation under actual operating conditions.

Verification coverage. All six illustrated H100 cases are actual learned CAD program edits, followed by execution and comparison with released STEP targets. Each is now submitted to a separate 3D FEM response and mesh audit under the documented synthetic contract. The online benchmark and localized mechanical pilot check geometry; the plate controls check clearances. The six-case supplement solves 3D linear-elastic FEM problems, while the separate truss controls solve axial FEM equilibrium. Geometry agreement does not establish stress, stiffness or safety. FEM for a mechanical part would require an explicitly specified material, unit system, load case, supports and a validated mesh/-solver setup, which are absent from these released edit pairs.

7. Limitations and conclusion

This is a small, single-seed code-conditioned CAD editing study. The online supervision comes from a synthetic mechanical benchmark re-partitioned locally; family/template checks do not prove complete geometric-ancestry isolation. A documented instruction/target conflict and one invalid STEP reference show that published labels require further auditing. Context filtering removes some of the longest parts and edit patches. Predicted code is limited by the executor and its time budget, and the current comparison checks

solid volume agreement rather than feature-specific engineering tolerances. Our SVG views are deterministic projections; no learned drawing perception, annotation binding or production-sheet reconstruction is evaluated.

The structural controls are deterministic and use bounded grammars. Their equilibrium and clearance results cannot be transferred to arbitrary CAD shapes or interpreted as general safety certification. The full mechanical benchmark has no author-specified physical contract. We add a separate post-hoc 3D FEM audit of six illustrated edits under documented synthetic fixtures and loads; its response checks do not establish operating safety.

The completed workflow links published CAD edit supervision, validated program patches, executed solids and vector views. By reporting schema adherence, target syntax, actual solid agreement and operational failures separately, it makes the supported outcome reviewable. Stronger evidence requires larger contexts, multiple seeds and models, feature-preservation metrics, genuine drawing-conditioned editing and physical tasks with independently specified analysis contracts.

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A. Reproduction and evidence contracts

Online source. The `edit-bench` parquet is downloaded from <https://huggingface.co/datasets/BenchCAD/BenchCAD> at the revision stated in Sec. 4. The original published split name is `edit_bench`. Its 748 rows are not silently treated as 748 independent source shapes; grouping uses all original and edited programs before split assignment. Numeric literals are masked for template linkage; all records of a linked component retain one split.

```
{ "edits": [
  { "start": 5, "delete": 1,
    "insert": [ "    .hole(6.0)" ] }
]
```

The zero-based line coordinate addresses the original code. A replacement list is required even for one line. Insertion uses zero deletions. The executor rejects overlapping operations and embedded newline characters. This example describes the interface, not an online gold answer.

Saved artifacts. The online dataset manifest pins the parquet checksum, split checksums, source IDs, categories and reference STEP hashes. The run manifest pins model revision, package versions, seed, optimization command, retained rows and omissions. The baseline and reloaded final adapter use identical test prompts. Predictions remain unmodified; the bare-list transport control is saved as a separate diagnostic. Geometry scores include every retained case, including invalid patches, code errors, unsupported constructs, timeouts and Boolean-comparison errors.

Evidence separation. The 150-pair localized pilot, 710-pair online protocol and historical 40,000-edit structural audit are different datasets and contracts. Their results are never pooled. The localized pilot’s family test contains only 12 context-retained examples and gives line/number hints. The online experiment is trained afresh with original natural-language requests and evaluates 126 context-retained cases. Structural FEM is only established for specified pin-jointed trusses.

Validation. Regression checks test multi-operation original-coordinate execution, insertion/deletion, malformed indexing, overlap rejection and comment preservation. Dataset checks require train/validation/test family and template-component disjointness, exact target-patch replay and pinned split hashes. Final adapters are reloaded for generation and checked

for finite tensors. Reference STEP readback and all output geometry scores are retained. Source/reference illustrations are labeled as such and do not depict model predictions.

B. Held-out model edit examples

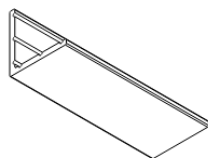
Figures 2 and 3 show six actual predictions from the final three-epoch H100 adapter, alongside the input solid and released STEP target. These are illustrative cases selected after evaluation, not a random sample or additional test set; all 126 evaluated cases remain included in the reported aggregate results. We select three strict matches and three executable but geometrically incorrect edits to expose both supported behavior and remaining limitations. Each view is independently fitted to its panel, so apparent drawing size is not a dimensional comparison. Instructions and measured volume IoU accompany every row. These shape plates alone provide no FEM evidence; the separate scenario-defined audit below supplies the material, loading, support and solver records.

Held-out H100 CAD edits: strict matches

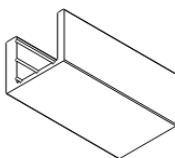
mounting angle box x f130 | T1 | IoU 1.00000

Increase the box X dimension from 33.0 to 42.9.

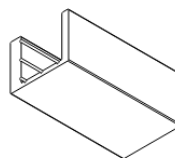
SOURCE



RELEASED TARGET



H100 PREDICTION



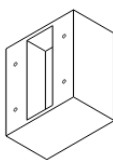
locator block box y f130 | T1 | IoU 1.00000

Increase the box Y dimension from 82.2 to 106.86.

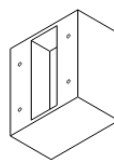
SOURCE



RELEASED TARGET



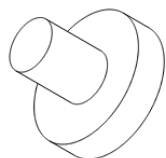
H100 PREDICTION



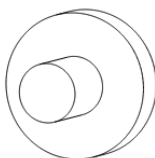
t3medplus stepped shaft rot diag60 | T2 | IoU 1.00000

Rotate the entire part by 60 degrees about the axis (1, 0, 1).

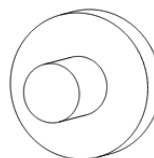
SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

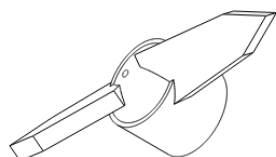
Figure 2. Three successful held-out H100 edits. Rows show a mounting-angle dimension change, locator-block width change and a 60-degree stepped-shaft rotation. Columns contain the source, author’s released target and actual final-adaptor prediction. Each meets strict volume IoU ≥ 0.99999 . These examples demonstrate reference agreement within the disclosed code-conditioned protocol; they do not certify general engineering correctness.

Held-out H100 CAD edits: partial / failed edits

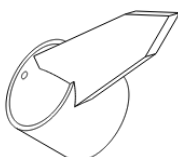
propeller bore widen compute | T4 | IoU 0.63155

Remove one of the two propeller blades; keep the hub, spinner, and the remaining blade.

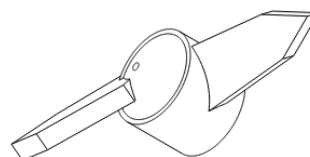
SOURCE



RELEASED TARGET



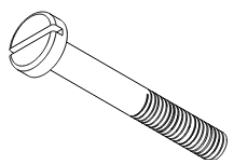
H100 PREDICTION



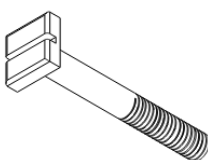
t5hard pan head screw U a | T5 | IoU 0.58449

Replace the round pan head of the screw with a square head of the same across-flats width and the same thickness, keeping the threaded shaft, the screwdriver slot, and the rounded top edge unchanged.

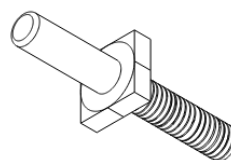
SOURCE



RELEASED TARGET



H100 PREDICTION



t5hard round flange V a | T5 | IoU 0.92018

Change the ring of 4 polar mounting through-passages on the flange face to a ring of 6 mounting through-passages of the same diameter, evenly spaced around the central axis at the same radius.

SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. FEM audit reported separately.

Figure 3. Three executable but incorrect held-out H100 edits. Removing a propeller blade, replacing a screw head and changing a flange mounting-hole pattern yield valid solids but fail the strict reference threshold. Their volume IoUs are approximately 0.632, 0.584 and 0.920. Source, released target and model output remain separate; no reference operations repair these predictions. Small feature errors can remain important even when bulk volume overlap is high.

C. Scenario-defined FEM audit of mechanical edits

Scope and physical contract. We attempt FEM on every source, released target and actual prediction in the six illustrated H100 cases: 18 configurations, independently of model training. The released dataset supplies no operating load case. We explicitly assume millimetres, newtons and MPa, a homogeneous isotropic material with $E = 210,000$ MPa and $\nu = 0.3$, and a 1 N resultant tensile test load. These are synthetic diagnostic conditions, not recovered material properties or certified operating conditions. All displacements are fixed at exterior mesh nodes in the lower 5 % of the shape’s projected span along the analysis axis. A uniform vector traction is integrated over exterior triangles whose centroids occupy the upper 5 %; its area-weighted nodal forces sum to 1 N. The same geometric fixture rule is applied separately to each shape, rather than claiming identical world-coordinate contact patches. The shaft’s fixture and load axis co-rotate with its requested 60-degree pose edit. Every case contract and STEP hash is saved before observing FEM responses.

Discretization and numerical checks. Gmsh [2] meshes the actual STEP geometry with first-order tetrahedra. A SciPy sparse implementation solves small-strain 3D isotropic elasticity,

$$\epsilon = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad \sigma = \lambda \operatorname{tr}(\epsilon) \mathbf{I} + 2\mu \epsilon. \quad (2)$$

The element stiffness is $K_e = V_e B_e^T D B_e$, with engineering shear strains. Dirichlet elimination solves the free-degree system. Checks require finite displacements, anchored connected components, positive strain energy, free residual, force and moment balance, and the identity $2U = \mathbf{f}^T \mathbf{u}$, each normalized numerical error below 10^{-6} . An analytical uniform-tension cube agrees to numerical precision; load/modulus scaling and co-rotated-fixture tests also pass.

Mesh sensitivity and response comparison. We report compliance $C = \mathbf{f}^T \mathbf{u}$ in Nmm. Global mesh sizes start at one twelfth and one eighteenth of the case’s maximum bounding-box extent, with further refinement if the relative compliance change exceeds 5 %. The propeller source requires a third mesh. A local target-response diagnostic accepts compliance discrepancy at most 10 %, but this does not replace strict solid agreement. Stress fields and volume-weighted 95th-percentile von Mises values are retained for inspection; compliance convergence does not certify stress convergence. Clamp singularities, small feature errors, yield,

buckling, fatigue, contact and actual-use safety are outside this experiment.

Five of six case triples yield 15 solved source/target/prediction configurations with compliance mesh changes below 5 %. Across their recorded meshes the largest normalized equilibrium/energy error is $8.20\text{e-}13$; final CAD-to-mesh volume error is at most 1.39 %. Three predictions meet both strict geometry agreement and the local 10 % target-compliance agreement tolerance. The propeller and flange predictions fail geometry while meeting that response tolerance. The screw has no FEM response because its boundary mesh fails; this also affects the source and released target, so it is not attributed solely to the model. The incorrect screw output contains seven CAD solids, versus one in its target. Boundary meshing encounters intersecting facets/segments; a source-only finer-mesh and alternative HXT-mesher diagnostic also fails. No component fusion, thread removal, contact model or geometry repair is used to manufacture a solution. This case remains not verified.

D. Multi-source Colab training pipeline

The reproducible extension uses paired edits from BenchCAD and Microsoft’s CAD-Editor in their native representations. The prepared split contains 2,540 training, 172 validation, and 252 test records: respectively 492/44/124 CadQuery pairs and 2,048/128/128 native sketch-extrude pairs. Two additional pipe-related held-out instructions are excluded from this extension; the earlier 126-case experiment is unchanged. Exact source/target connected components are isolated within each source. Cross-source geometric ancestry is not established. Token-length filtering is recorded before training and evaluation.

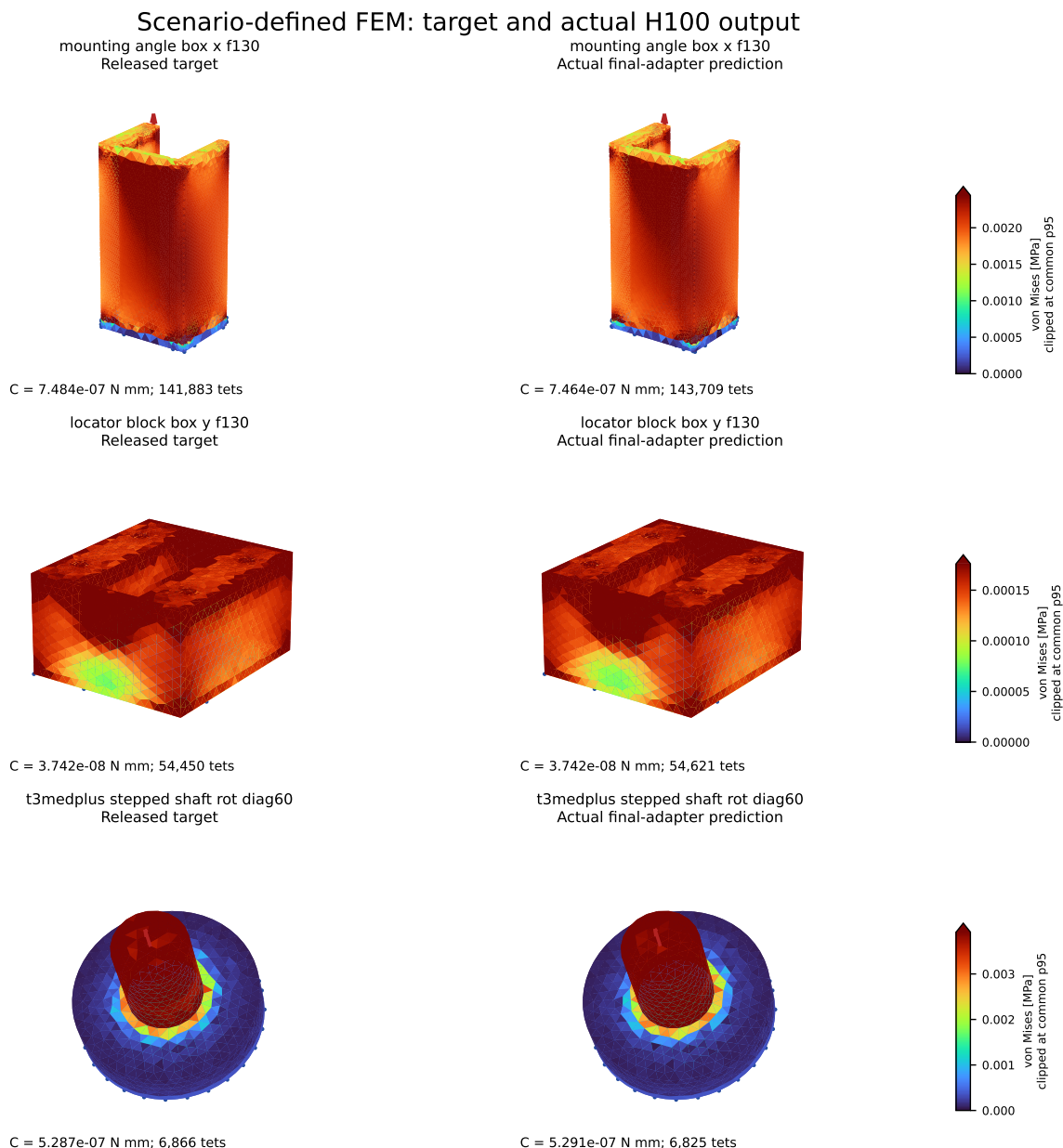
A fresh Qwen2.5-Coder-3B-Instruct backbone uses NF4 double quantization and completion-only QLoRA: rank 16, alpha 320, the seven attention/MLP projections in the last twelve blocks, two fixed epochs, seed 17, learning rate 10^{-4} , effective batch four, a 4,096-token context, and a 768-token output cap. Colab T4 computation uses FP16 and microbatch one with four accumulation steps. The final adapter is reloaded for greedy evaluation. This extension is an execution protocol in progress; no completed multi-source result is asserted here.

The model maps an instruction and original native CAD to an original-coordinate JSON line patch. Representation-specific schema and syntax checks precede execution. CadQuery patches are compared to released STEP targets; CAD-Editor command patches execute through a six-bit sketch-extrude decoder, with locally reconstructed target solids explicitly distin-

Example	Source C	Target C	Prediction C	Discrepancy (%)	Combined diagnostic
Mounting angle	2.043e-06	7.484e-07	7.464e-07	0.259	Geometry + response agree
Locator block	5.185e-08	3.742e-08	3.742e-08	0.003	Geometry + response agree
Rotated shaft	5.287e-07	5.287e-07	5.291e-07	0.068	Geometry + response agree
Propeller	3.334e-07	5.242e-07	5.221e-07	0.409	Geometry fails; response agrees
Pan-head screw	—	—	—	—	Not verified: mesh failure
Round flange	1.208e-07	1.212e-07	1.212e-07	0.021	Geometry fails; response agrees

Table 5. Mechanical example FEM checks under explicit test scenarios. Compliance values use N mm. Discrepancy compares the prediction with the target, with independently generated meshes. A response match is not sufficient for a correct edit. Dashes denote failed meshing, not zero displacement or an invented solver pass.

guished from author-released STEP files. Geometry and failure counts remain separate by source. Only examples with explicit material, unit, fixture, and load contracts proceed to tetrahedral FEM. Equilibrium, energy identity, mesh-volume fidelity, and compliance sensitivity checks precede a combined geometry/response diagnostic. FEM is an external post-edit tool, not a language-generated pass label or a solver reward used in training. Invalid patches and failed meshes remain unverified. The notebook saves adapters, manifests, predictions, geometry metrics, and numerical fields before downloading an artifact bundle.



Assumed $E = 210,000$ MPa, $\nu = 0.3$; 1 N test load. Blue: fixed nodes; red: load direction.
Undeformed meshes; compliance mesh check only. Stress fields are diagnostic, not a strength/safety verdict.

Figure 4. Actual FEM fields for three geometry-matching H100 edits. Released targets and model outputs use the documented synthetic test contracts. Blue points identify fixed nodes; arrows indicate the load direction. Undeformed surface stresses share a scale within each row, clipped above the common 95th-percentile value. Compliance passes the mesh-sensitivity check; stress colors are diagnostic rather than a strength verdict.

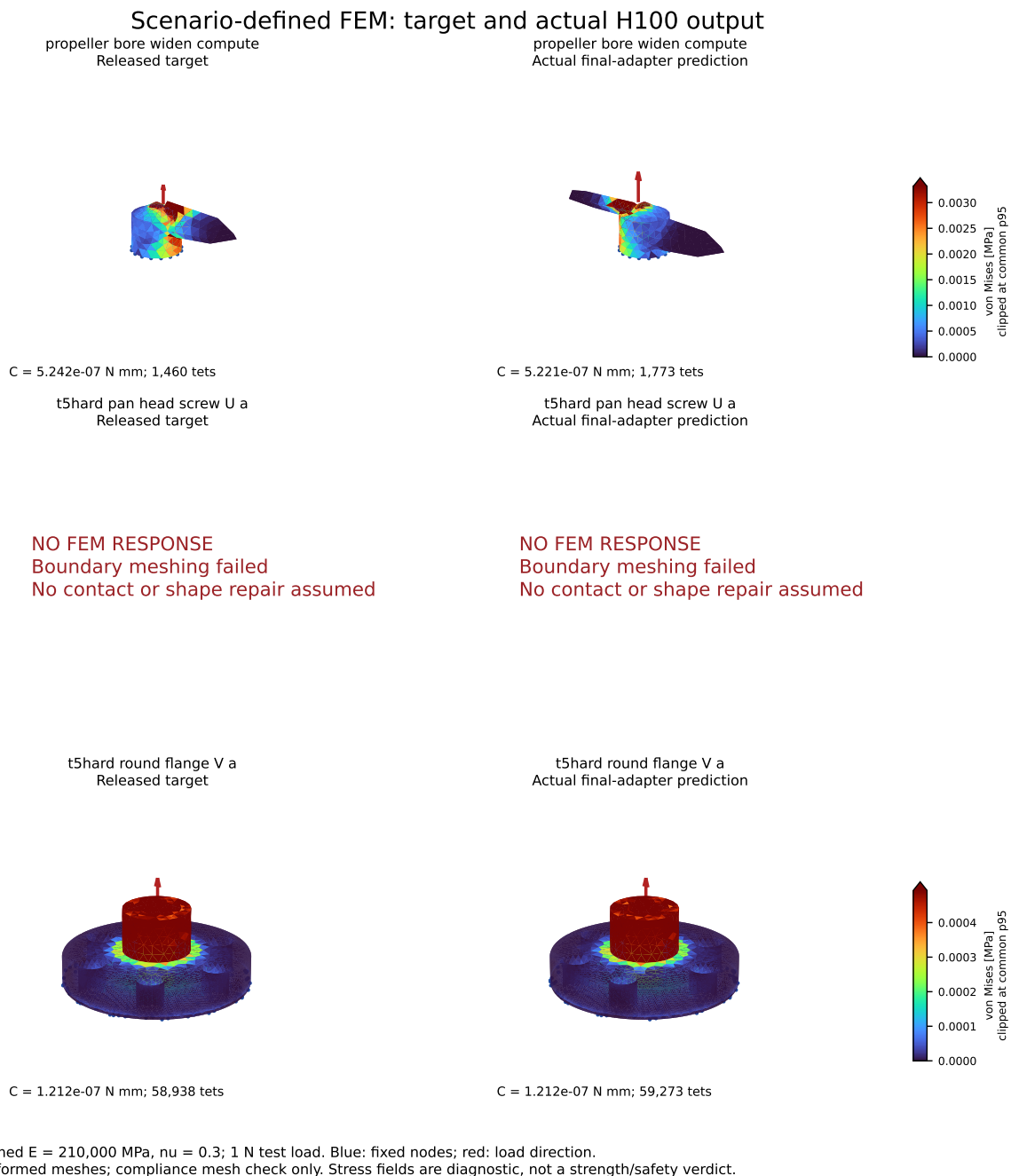


Figure 5. FEM diagnostics do not replace edit fidelity. Incorrect propeller and flange shapes produce similar compliance under the chosen test load, but fail strict reference geometry. The screw row has no FEM field because meshing fails; blank panels are explicitly marked. The same stress-scale and fixture conventions apply as in Fig. 4.